Toward Visually Robust Quadruped Parkour via Monocular Depth Estimation

Theo Lemay
Undergraduate Thesis
theo.lemay@queensu.ca

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Abstract

Vision-based quadruped parkour typically relies on onboard depth cameras, limiting deployment to robots with specialized sensors. We investigate whether a pre-trained *monocular depth estimator* can replace depth cameras for locomotion by producing a canonical depth representation from RGB, without photorealistic rendering or domain randomization.

We present a two-stage pipeline: a privileged teacher trained via PPO with ground-truth terrain scans, distilled into a visual student via DAgger operating on depth images from Depth Anything V2 (DA2). In simulation, this pipeline enables basic parkour on stairs, hurdles, and gaps under conditions characterized by a *Depth Signal Quality* (DSQ) metric. Visual robustness experiments show that DA2-based students tolerate moderate lighting and color perturbations within simulation, though performance degrades on featureless surfaces and under combined perturbations. All results are in simulation; real-world transfer remains future work.

1 Introduction

Legged robots that traverse irregular terrain—stairs, hurdles, and gaps—must perceive geometry and coordinate vision and control at high rates. Training locomotion policies in simulation with reinforcement learning (RL) and transferring them to real hardware has proven effective for agile skills [8, 11, 15]. Recent work [2, 4, 7, 24] has extended this approach to parkour—climbing, jumping, and crawling—through teacher–student distillation. In these systems, a *teacher* trains with privileged terrain information (e.g. ground-truth heightmaps), and a *student* reproduces the teacher's behavior from onboard sensors.

A central assumption, however, is access to an onboard depth sensor. Depth-camera policies transfer well from simulation on platforms that provide depth sensing [1, 12], but not all robots do. The Unitree Go2, for instance, ships with RGB cameras and a LiDAR sensor, but no onboard depth camera [20].

Deploying vision policies from RGB therefore requires bridging the *sim-to-real gap*. Three strategies have emerged: (1) **domain randomization** [19], which randomizes visual properties; (2) **generative augmentation** [23], which uses diffusion models for photorealistic training images; and (3) **canonical representations**, which project both domains into a shared space via a foundation model.

We pursue strategy (3) using Depth Anything V2 (DA2) [22], a pre-trained monocular depth estimator trained on millions of diverse real images. Our premise is that if DA2 maps RGB into a depth space that preserves terrain geometry across visual conditions, then a student policy operating on estimated depth may inherit partial robustness to appearance variation without photorealistic rendering or explicit domain randomization.

Thesis statement. We hypothesize that pre-trained monocular depth estimation can provide a useful intermediate representation for vision-based quadruped parkour, enabling a student trained on estimated depth in simulation to tolerate a range of visual conditions without explicit domain randomization. We evaluate this hypothesis entirely in simulation; real-world transfer remains future work.

Contributions.

- A two-stage teacher–student pipeline from privileged scandot heightmaps to monocular depth for quadruped parkour.
- A demonstration that estimated depth enables basic parkour in simulation.
- Identification of depth signal quality—governed by model scale and terrain texture—as the key factor limiting performance.
- A visual robustness study showing that DA2-based students tolerate lighting and color shifts but depend on terrain texture for depth signal quality.

2 Related Work

RL for legged locomotion. Modern quadruped controllers are typically learned via reinforcement learning in simulation and transferred to real hardware [8, 15]. Chen et al. [3] introduced the privileged teacher–student paradigm, where a teacher with access to ground-truth state is distilled into a sensorimotor student. Lee et al. [11] applied this to locomotion, training a student from proprioceptive history alone and achieving zero-shot sim-to-real transfer on rough terrain. Kumar et al. [10] introduced rapid motor adaptation, using a learned adaptation module to estimate privileged parameters online. Miki et al. [12] extended this framework to perceptive locomotion, incorporating exteroceptive depth inputs.

Vision-based parkour. Agarwal et al. [1] introduced the scandot representation for egocentric depth-based locomotion, distilling a privileged teacher into a visual student. Extreme Parkour [4] and Robot Parkour Learning [24] build on this with CNN–GRU students that process depth images, demonstrating climbing, jumping, and crawling on real quadrupeds. SoloParkour [2] replaces distillation with off-policy RL warm-started from privileged experience, and ANYmal Parkour [7] uses a hierarchical skill-selection policy. All five systems assume access to a depth camera at deployment.

Jenkins [9] is the closest work to ours, training an RGB parkour policy using visual domain randomization with a fine-tuned MobileNetV2 backbone. In real-world trials, Jenkins succeeded on hurdles but failed on gaps, suggesting that domain randomization alone may not fully close the visual sim-to-real gap. We pursue a complementary strategy: instead of learning invariance through randomized exposure, we project RGB into a canonical depth space using a pre-trained estimator.

Sim-to-real for vision. Domain randomization [19] perturbs visual properties (textures, lighting, colors) so the policy learns invariance through exposure. LucidSim [23] replaces randomization with generative models that produce photorealistic training images. Xue et al. [21] combine photorealistic simulation with a teacher–student–bootstrap framework for humanoid loco-manipulation from RGB. All three approaches require substantial rendering or generation overhead. We instead project RGB into a canonical depth space using a pre-trained estimator.

Monocular depth estimation. MiDaS [13] pioneered monocular depth estimation across diverse datasets. Depth Anything V2 (DA2) [22] achieves state-of-the-art zero-shot depth estimation using a vision transformer encoder trained on 142M images. DA2 offers both relative and metric variants; we use the metric model, which predicts absolute depth in meters. Figure 1 illustrates the qualitative behavior of DA2 across diverse scenes and motivates its use here as a canonical visual representation.

Figure 1: Depth Anything V2 [5, 22].

3 Mathematical Foundations

3.1 Problem Formulation

We model the parkour task as a Partially Observable Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, observation space $\mathcal{O}$, transition dynamics T, reward function R, and discount factor γ. The agent receives partial observation $o_t \in \mathcal{O}$ and selects action $a_t \in \mathcal{A} = \mathbb{R}^{12}$ (target joint position offsets). Observations decompose into three groups:

- **Proprioception** $o_{\text{prop}} \in \mathbb{R}^{53}$: base angular velocity, projected gravity, velocity and heading commands, joint positions, joint velocities, previous actions, and foot contacts.
- **Exteroception** o_{ext}: scandot heightmaps ($\mathbb{R}^{132}$) for the teacher, or depth images for the student. Scandots [1] are terrain height samples on a 12×11 grid in the robot's body frame. Each value records the height difference between the base and the terrain surface, clipped to $[-1, 1]$.
- **Privileged** o_{priv}: friction and other physics parameters available only during teacher training.

3.2 PPO

The teacher policy π_θ is trained with Proximal Policy Optimization (PPO) [17], which maximizes a clipped surrogate objective:

$$L^{\text{CLIP}} = \mathbb{E}_t\Big[\min\Big(r_t\hat{A}_t,\ \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\,\hat{A}_t\Big)\Big], \quad (1)$$

where $r_t = \pi_\theta(a_t|o_t)/\pi_{\theta_{\text{old}}}(a_t|o_t)$ is the probability ratio and $\hat{A}_t$ is the GAE [16] advantage estimate.

3.3 Reward Function

The teacher uses a shaped reward that combines task progress with regularization for stable locomotion. Positive terms reward tracking the commanded velocity and heading, making forward progress, and reaching waypoints and course completion. Penalty terms discourage early termination, collisions, unstable body motion, large torques and joint accelerations, abrupt action changes, and joint or torque limit violations. See the full reward function in Table 4.

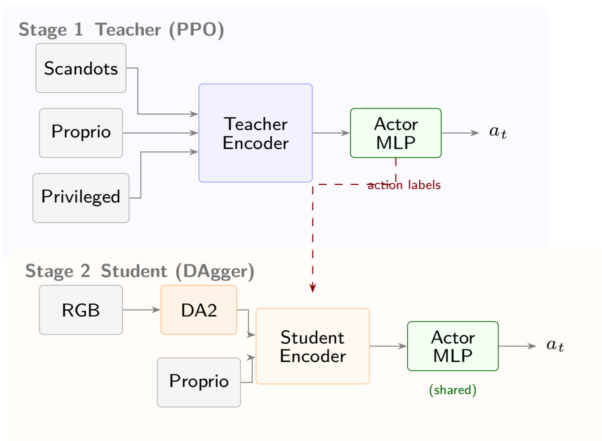

Figure 2: Two-stage training pipeline. The teacher uses privileged scandot observations; the student replaces these with either GT depth or DA2-inferred depth.

3.4 DAgger Distillation

The student policy is trained via Dataset Aggregation (DAgger) [14]. At each step the student observes o_{prop} and a depth image and predicts a_{student}, but the teacher's action a_{teacher} is executed in the environment. The student minimizes:

$$\mathcal{L} = \underbrace{\|a_{\text{teacher}} - a_{\text{student}}\|_2}_{\text{action loss}} + \underbrace{\|y_{\text{oracle}} - \hat{y}\|_2}_{\text{yaw loss}}, \tag{2}$$

where $\hat{y} \in \mathbb{R}^2$ is the student's predicted heading and y_{oracle} is the ground-truth heading.

4 System Architecture

4.1 Overview

All experiments use the Genesis GPU simulator [6] with the legged_gym framework [15] and rsl_rl [18] on 4× NVIDIA L40S GPUs. The robot is a Unitree Go2 quadruped with 12 actuated joints. Actions $a_t \in \mathbb{R}^{12}$ are target joint positions, updated at 50 Hz and converted to torques by a PD controller $\tau = K_p(a_t - q) - K_d\dot{q}$.

The system follows the two-stage privileged distillation pipeline described in Section 1: a teacher trains with ground-truth scandot heightmaps [1], then a student reproduces the teacher's behavior from GT depth or DA2-estimated depth via DAgger (Figure 2).

4.2 Teacher

The teacher policy π_T maps proprioception, privileged physics parameters, and scandot heightmaps to joint targets. Three parallel MLPs encode the observation groups into 32-dim latent vectors, which are concatenated with proprioception and fed through the actor MLP $\mathbb{R}^n \to \mathbb{R}^{12}$ to produce joint position offsets.

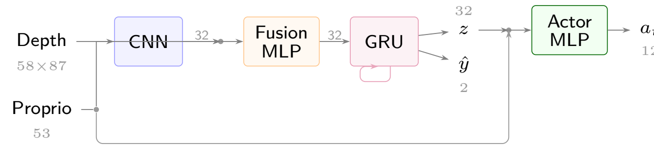

Figure 3: Student visual encoder. The depth CNN and proprioception are fused and processed by a GRU that produces the terrain latent and heading estimate used by the teacher's actor MLP.

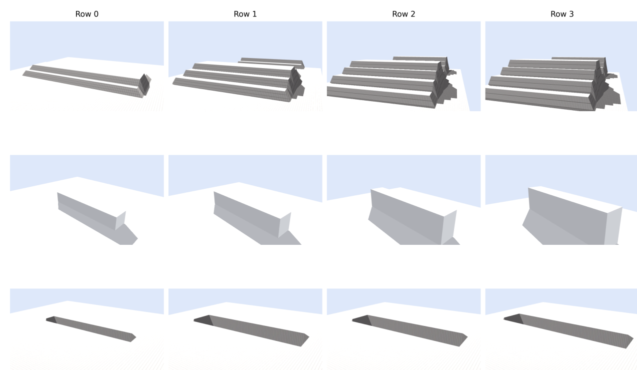

Figure 4: Parkour curriculum progression across obstacle families. Difficulty increases from left to right by increasing step count/height, hurdle size, and gap width.

4.3 Student

The student replaces the scandot encoder with a visual encoder that recovers the terrain latent $z \in \mathbb{R}^{32}$ from a depth image (Figure 3). A CNN maps the depth image $d \in \mathbb{R}^{58\times 87}$ to a 32-dim feature, which is concatenated with proprioception and compressed by an MLP. A GRU integrates these fused features across timesteps, producing z and a heading estimate $\hat{y} \in \mathbb{R}^2$. The teacher's actor MLP is reused, taking $[o_{\text{prop}}; z]$ to output joint targets. During DAgger distillation, the student minimizes the loss in Eq. 2 while the teacher's actions are executed in the environment.

4.4 Terrain and Curriculum

The simulator generates parkour terrains with three obstacle families (stairs, hurdles, gaps) at four difficulty levels per family (Figure 4) with gap widths from 0.24 m to 0.50 m. These obstacles are simpler than those in prior work [2, 4, 24]; our goal is to establish a baseline for evaluating the DA2 depth pipeline rather than to push the limits of agile locomotion. During teacher training, a curriculum advances robots to harder rows upon reaching a success threshold.

Table 1: Baseline success rates on gap terrain (%).

	Row 0	Row 1	Row 2	Row 3
Scandots teacher	98.5	100.0	99.0	98.5
GT-depth student	94.5	94.5	89.1	87.5
Zero-depth ablation	8.0	2.0	0.0	0.0

5 Experiments and Results

We evaluate the DA2 pipeline through three experiments: (1) establishing upper-bound baselines with ground-truth sensing, (2) closing the gap between estimated and ground-truth depth by varying model capacity and terrain texture, and (3) testing visual robustness under visual perturbations. We report *success rate* (fraction of episodes completing the full course) and *progress ratio* (mean fractional distance along the course before termination), both averaged over 256 episodes.

For visual robustness, we also report *depth signal quality* (DSQ), defined as the scale- and offset-invariant Pearson correlation between ground-truth and DA2-estimated depth images, averaged over environments and time steps.

$$\mathrm{DSQ} = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{N}\sum_{i=1}^{N}\frac{\sum_p (d_{i,p}^{\mathrm{GT}} - \bar{d}_i^{\mathrm{GT}})(d_{i,p}^{\mathrm{DA2}} - \bar{d}_i^{\mathrm{DA2}})}{\|d_i^{\mathrm{GT}} - \bar{d}_i^{\mathrm{GT}}\|_2 \, \|d_i^{\mathrm{DA2}} - \bar{d}_i^{\mathrm{DA2}}\|_2} \tag{3}$$

5.1 Baselines

We first establish upper bounds for the privileged teacher and the ground-truth depth sensing student (GT-depth). The privileged teacher achieves near-perfect success across all gap difficulties, shown in Table 1.

The GT-depth student, receiving simulator depth images, achieves success on all rows, confirming the CNN–GRU architecture extracts sufficient geometry from a single depth viewpoint. A zero-depth ablation (all depth pixels set to zero) drops success to 8%, verifying the policy relies on depth rather than exploiting proprioceptive heuristics.

5.2 Parkour from Estimated Depth

We next test whether parkour is possible from estimated depth. Starting from the GT-depth setup, we replace simulator depth with DA2-small. As shown in Table 2, DA2-small succeeds on the early rows but fails completely on Rows 2 and 3. Switching to DA2-base reaches Row 3, but still falls well short of GT-depth on the hardest rows.

The remaining limitation is the visual input. On untextured terrain, the gap boundary provides weak cues, so the depth estimate is ambiguous. As shown in Table 2, adding a checkerboard terrain texture makes the gap geometry visible and closes most of the remaining gap to GT-depth. Figure 6 shows examples of depth estimation across representative surface conditions.

Table 2: Policy success rate (%) as model scale and terrain texture vary.

Configuration	Row 0	Row 1	Row 2	Row 3
DA2-small	56.3	33.6	—	—
DA2-base	68.4	83.2	26.6	0.8
DA2-base + texture	87.5	93.0	87.1	80.5
GT-depth baseline	94.5	94.5	89.1	87.5

Figure 5 shows the training dynamics. Reward rises steeply as the policies master flat terrain, then plateaus as the curriculum exposes harder obstacles. The textured model surpasses the untextured models and approaches the GT-depth baseline.

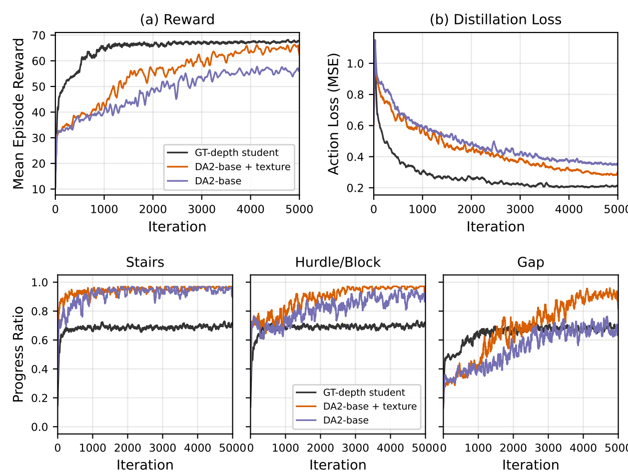

Figure 5: Success rate during student training. Breaking success down by obstacle type shows that most of the gain comes from gap traversal; stairs and hurdles are less sensitive to texture because their geometry remains visible.

5.3 Visual Robustness

We next test robustness to the appearance shifts that would arise in sim-to-real transfer by perturbing the trained student only at *test time*. All perturbations are applied within simulation; these results therefore measure within-sim robustness, not actual sim-to-real transfer. Table 3 reports success across six terrain configurations (three obstacle families × two difficulty levels) for texture replacement, lighting/color changes, and combined perturbations, while Figure 7 plots DSQ against policy success.

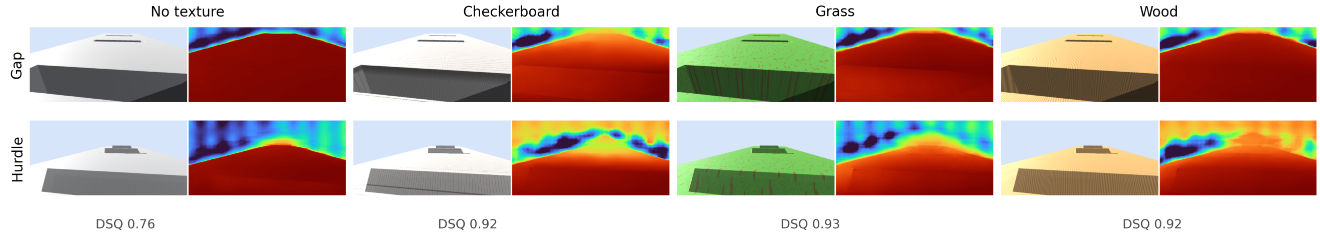

Figure 6: Robot-view comparison across representative terrain textures. Each panel shows rendered RGB (left) and the corresponding DA2 depth estimate (right). Featureless terrain produces the weakest obstacle localization and the lowest DSQ, while checkerboard, grass, and wood all preserve substantially more geometric structure.

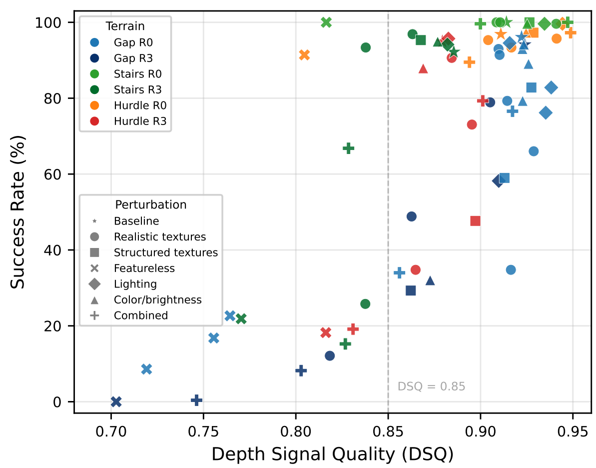

Figure 7: Success rate vs. DSQ across all six terrain configurations. Color denotes obstacle type and difficulty; marker shape denotes perturbation category. Featureless surfaces cluster at low DSQ with catastrophic failure; most textured conditions maintain DSQ > 0.85 but success varies by difficulty.

Table 3: Visual robustness: test-time success (%) under visual perturbations across obstacle types.

	Gaps		Stairs		Hurdles	
Perturbation	R0	R3	R0	R3	R0	R3
Baseline	96.1	94.1	100	92.2	96.9	94.9
Realistic surface textures						
Concrete	79.3	12.1	99.6	25.8	95.7	73.0
Stone tiles	93.0	78.9	100	96.9	95.3	90.6
Gravel	91.4	48.8	100	93.4	93.4	34.8
Structured / featureless						
Noise	82.8	29.3	100	95.3	97.3	47.7
No texture	16.8	0.0	100	21.9	91.4	18.2
Brightness / color						
Dim (0.3×)	82.8	58.2	99.6	94.1	99.6	95.7
Color ±30%	79.3	32.0	99.6	94.9	97.3	87.9
Combined[*]						
Mild	76.6	8.2	100	15.2	97.3	79.3
Extreme	34.0	0.4	99.6	66.8	89.5	19.1

[*] Mild: concrete + dim (0.6×) + brightness jitter ±20%. Extreme: noise + dim (0.4×) + low contrast (γ=1.8) + color jitter ±20%.

The results reveal a clear obstacle-geometry hierarchy. At the easier setting, stairs and hurdles are largely insensitive to appearance changes because strong geometric edges remain visible across textures and lighting. Gaps are more sensitive because their boundary is visible mainly through surface texture, so performance depends more strongly on the quality of the depth estimate.

At the harder setting, all three obstacle families degrade, including stairs and hurdles. Featureless surfaces cause the largest failures, while lighting and color shifts have a smaller effect. This pattern suggests DA2 is robust to photometric variation but still depends on texture when obstacle boundaries are weak.

As Figure 7 shows, DSQ is a useful failure indicator: featureless surfaces collapse both DSQ and success. However, high DSQ alone does not guarantee robustness, since some unseen textures still reduce success despite preserving coarse depth structure.

6 Discussion and Conclusion

6.1 Summary

We presented a teacher–student pipeline for vision-based quadruped parkour that uses monocular depth estimation as an intermediate representation. The GT-depth baseline shows that the student architecture can solve the task when accurate geometry is available. The DA2 experiments show that the main bottleneck is the quality of the recovered depth signal rather than the policy architecture. Increasing model capacity and adding structured terrain texture closes most of the gap to the GT-depth baseline on the gap curriculum.

The visual robustness experiments further refine this result. The DA2-based student tolerates moderate lighting and color changes and transfers to several unseen textured surfaces without photorealistic rendering or explicit domain randomization. But the method is not appearance-invariant in general: performance degrades on harder settings and collapses on featureless surfaces, where monocular depth cannot recover reliable geometry. DSQ is a useful indicator of whether the visual pipeline preserves terrain structure, but it does not fully determine policy success. Overall, the results support monocular depth as a plausible bridge from RGB to locomotion, but actual sim-to-real transfer remains unvalidated.

6.2 Limitations

Several limitations constrain this conclusion. First, all experiments are in simulation, so the central sim-to-real claim is still unverified on hardware. Second, the task scope is modest: the obstacles are simpler than those in recent real-world parkour systems, and the visual robustness study covers two difficulty levels across three families rather than the full curriculum. Third, the method depends on a single off-the-shelf depth estimator. DA2 degrades on weak-texture terrain, and high DSQ does not explain all failures on unseen textures. Fourth, DA2 processes each frame independently, so depth estimates may flicker between consecutive timesteps. The GRU in the student encoder may partially smooth such temporal noise, but we did not measure frame-to-frame depth consistency or test temporally consistent depth models, leaving this an open question for deployment at high control rates. Fifth, our visual robustness study covers texture, lighting, and color perturbations, but real-world deployment introduces additional failure modes that monocular depth estimators are known to struggle with: transparent or reflective surfaces, thin structures such as railings, motion blur during dynamic gaits, and close-range objects at the limits of the model's training distribution. These are likely to arise in outdoor parkour environments and may compound the texture sensitivity already observed. Finally, the perception stack is still heavy for onboard deployment. The current system is therefore best viewed as a proof of concept rather than a deployable parkour pipeline.

6.3 Future Work

The next step is real-world evaluation on the Unitree Go2. This will test robustness under real camera noise, motion blur, exposure shifts, and outdoor lighting, and determine whether DSQ remains useful as a deployment diagnostic. A second direction is broader evaluation across stairs, hurdles, and more difficult compound courses. On the modeling side, smaller or temporally consistent depth estimators could reduce compute cost and stabilize predictions, while joint adaptation of the depth estimator and student policy may improve robustness on harder textures and further close the gap to the GT-depth baseline.

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A Hyperparameters

This appendix lists the training hyperparameters used in the experiments. Unless noted otherwise, values are shared across all runs.

A.1 Reward Function

Table 4 lists the reward terms and scale factors used for teacher training. The reward at each step is $r_t = \sum_i w_i \, r_i(s_t, a_t)$, where w_i is the scale and r_i is the per-term reward signal.

Table 4: Teacher reward terms and scales.

Term	Scale
Task progress	
Tracking goal velocity	3.0
Tracking goal heading	1.5
Progress along course	2.0
Waypoint reached bonus	20.0
Success bonus	40.0
Termination & collision	
Termination	−50.0
Collision	−2.0
Feet stumble	−1.0
Feet edge	−2.0
Locomotion regularization	
Torques	-2.0×10^{-4}
Joint power	-2.0×10^{-4}
Joint acceleration	-2.0×10^{-7}
Action rate	−0.01
Action smoothness	−0.01
Joint position limits	−2.0
Torque limits	−0.2
Body stability	
Linear velocity z	−0.5
Angular velocity xy	−0.05
Orientation	−0.5
Flat back	−3.0

A.2 Training Hyperparameters

PPO and DAgger training hyperparameters are listed in Tables 5 and 6, respectively. Control parameters and domain randomization ranges are listed in Table 7.

Table 5: PPO hyperparameters for teacher training.

Parameter	Value
Environments	4096
Steps per env per update	48
Mini-batches	4
Learning epochs per update	5
Learning rate	1×10^{-3}
LR schedule	adaptive (target KL)
Desired KL	0.01
Discount γ	0.99
GAE λ	0.95
Clip ϵ	0.2
Entropy coefficient	0.01
Value loss coefficient	1.0
Max gradient norm	1.0

Table 6: DAgger hyperparameters for student distillation.

Parameter	Value
Environments	48
Steps per env per update	120
BPTT window	24
Learning rate	2×10^{-3}
Actor LR scale (DA2 student)	0.1
LR schedule (DA2 student)	cosine
Action loss coefficient	1.0
Yaw loss coefficient	1.0
Yaw threshold (MTS)	0.6
Max gradient norm	1.0

Table 7: PD control and domain randomization parameters.

Parameter	Value
PD controller	
K_p	30.0
K_d	0.75
Action scale	0.25
Control dt	0.02 s (50 Hz)
Domain randomization	
Friction range	[0.2, 1.7]
Added base mass	[−1.0, 1.0] kg
CoM displacement xyz	±0.03 m
K_p / K_d scale	[0.8, 1.2]
Push interval	15 s
Max push velocity	1.0 m/s